

MANUFACTURING TECHNOLOGY

Manufacturing is the transformation of materials into products to meet human needs and wants. Manufacturing technology courses provide students with opportunities to acquire knowledge and skills in the areas of mechanical engineering, robotics and control systems, computer-aided manufacturing, industrial maintenance, precision machining, welding, and sheet metal.

Students will use a broad range of tools and equipment while acquiring engineering, fabrication, and problem-solving skills through the design and fabrication of various projects. Manufacturing courses will also help students understand the effects of manufacturing technology on the environment and society.

Manufacturing courses are designed for students heading to an apprenticeship, to college or university, or directly to the workplace in their pursuit of careers such as tradesperson, technician, technologist, engineer, or any of the skilled entry positions.

The list of approved emphasis areas for manufacturing technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Manufacturing Technology, Grade 12

Workplace Preparation

TMJ4E

This project-driven, hands-on course builds on students' experiences in manufacturing technology. Students will further develop knowledge and skills related to the use of engine lathes, milling machines, welding machines, and other tools and equipment as they design and fabricate solutions to a variety of technological challenges in manufacturing. Students may also have opportunities to acquire industry-standard training and certification. Students will expand their awareness of environmental and societal issues and of career opportunities in the manufacturing industry.

Prerequisite: Manufacturing Technology, Grade 11, Workplace Preparation

A. MANUFACTURING TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of the secondary manufacturing industries and the processes and technologies related to them;
- A2.** demonstrate a working knowledge of how a design process is used to develop and fabricate projects in response to challenges or problems in manufacturing technology;
- A3.** demonstrate a working knowledge of the processes required for making material conversions;
- A4.** apply relevant mathematical and scientific concepts and technological literacy and communication skills in the study of manufacturing technology.

SPECIFIC EXPECTATIONS

A1. Secondary Manufacturing Industries

By the end of this course, students will:

- A1.1** describe how materials from secondary manufacturing industries can be used to produce finished products (*e.g., types of structural steel, hollow structural steel, wide flange steel, angle iron*);
- A1.2** identify and describe the key areas of operation in secondary manufacturing industries (*e.g., product design, production planning, product fabrication, quality control, worker safety*);
- A1.3** identify and describe production methods (*e.g., custom, assembly line, flexible, mass production*) used in the manufacturing of various products;
- A1.4** identify organizations that develop standards regarding the conception and manufacture of consumer goods (*e.g., Canadian Standards Association [CSA], International Organization for Standardization [ISO]*);
- A1.5** describe in detail the function, purpose, and operation of advanced machine tools and equipment (*e.g., computer numerical control [CNC] machines, water jet cutting machine, laser equipment, hydro-forming machine*) used in secondary manufacturing.

A2. Design Process

By the end of this course, students will:

- A2.1** demonstrate how a design process can be used in all phases of project development from conception to finished product (*e.g., in analysing the situation, defining the problem or need, synthesizing ideas, planning procedures, generating solutions, evaluating results for continuous product improvement*);
- A2.2** describe how the design process is used to improve manufacturing methods (*e.g., with respect to material flow, equipment layout, tool organization and maintenance, cost savings, efficiency*);
- A2.3** demonstrate appropriate use of various forms of communication (*e.g., brainstorming, discussion, presentations*), graphic representation (*e.g., sketches, technical drawings, computer-aided design [CAD]*), and documentation (*e.g., research reports*) in the development of a project;
- A2.4** demonstrate skills and attitudes needed to work as a cooperative member and/or leader of a group during the design process (*e.g., effective time management, goal setting, conflict resolution, communication skills, respect for differences, evaluation of strengths and weaknesses, ability to delegate tasks*);

A2.5 demonstrate a working knowledge of ways in which technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) are important considerations in the design process (see pp. 7–8).

A3. Material Conversion

By the end of this course, students will:

A3.1 apply appropriately a variety of processes used to join materials (*e.g., gas tungsten arc welding [GTAW], gas metal arc welding [GMAW], shielded metal arc welding [SMAW], bonding, fastening*);

A3.2 demonstrate a working knowledge of the processes used to cut or separate materials (*e.g., using a CNC machine, oxy-acetylene cutting torch, lathe, mill*);

A3.3 demonstrate correct processes for forming materials, using various tools and equipment (*e.g., brake press; English wheel; forging, casting, and moulding tools and equipment*);

A3.4 demonstrate correct processes for converting the structure of a material using a variety of metallurgical processes (*e.g., heat treating, annealing, shrinking, tempering/work hardening*).

A4. Mathematics, Science, and Technological Literacy

By the end of this course, students will:

A4.1 demonstrate appropriate use of various mathematical formulas (*e.g., related to trigonometry, sine rule, taper calculation, bend allowances*) and applications (*e.g., tap drill sizing, calculating machine speed and feed rate, indexing*) that are commonly used in manufacturing;

A4.2 use appropriate mathematical solutions to solve manufacturing problems (*e.g., use algebraic equations; calculate perimeters, volumes, areas*), consulting appropriate resources (*e.g., tables, charts*);

A4.3 apply mathematical concepts and scientific principles (*e.g., concept: torque; principle: Bernoulli's principle*) related to power transmission and mechanical mechanisms (*e.g., hydraulics, pneumatics, gears, pulleys, levers, couplings*);

A4.4 demonstrate a working knowledge of physical and mechanical properties of materials (*e.g., physical: appearance, density, surface texture; mechanical: hardness, malleability, ductility, elasticity*) that affect the suitability of materials for use in manufacturing;

A4.5 use appropriate manufacturing terminology to communicate information related to the fabrication of a project;

A4.6 demonstrate effective use of research skills (*e.g., obtaining and evaluating information from technical documents and manufacturing resources*) and technological literacy and communication skills (*e.g., log note writing, record keeping, data recording, memo writing, schedule creation*) to plan and document work on a manufacturing project.

B. MANUFACTURING TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use technical drawing skills and a design process to create engineering drawings that provide solutions to project development challenges;
- B2.** select and use the proper materials when manufacturing a product to meet specifications;
- B3.** apply advanced metrology skills to measure, lay out, and inspect a product;
- B4.** demonstrate the safe and proper use of tools and equipment when producing various projects to meet specifications.

SPECIFIC EXPECTATIONS

B1. Technical Drawings and Process Plans

By the end of this course, students will:

- B1.1** demonstrate a working knowledge of how the design process is used to develop an effective solution to a manufacturing problem or challenge (*e.g., describe and analyse a situation and define the specific problem or need; research criteria, constraints, and availability of materials; generate a number of possible solutions, using techniques such as brainstorming; select the optimal solution; test and evaluate the product; repeat steps as necessary*);
- B1.2** demonstrate the ability to accurately interpret detailed engineering drawings and assembly drawings (*e.g., with respect to welding symbols, legends, dimensions, tolerances*) related to the development of a project;
- B1.3** interpret and/or create sketches or drawings (both manual and computer-aided) that include the measurements and symbols required to complete a project;
- B1.4** select appropriate resources and reference materials (*e.g., charts, tables, material- and product-related specifications, schematics*) when planning a project.

B2. Selection and Use of Materials

By the end of this course, students will:

- B2.1** select, organize, and prepare appropriate materials for the manufacture of a product

based on material properties (*e.g., physical, mechanical, thermal*) and design criteria (*e.g., hot or cold rolled steel, size and grade of steel drill rod, carbon content, ferrous or non-ferrous metals, type of welding electrode and/or gas*);

- B2.2** select appropriate materials for a project based on identified criteria such as functionality, cost effectiveness, customer expectations, viability, availability, and sustainability;
- B2.3** demonstrate a working knowledge of metallurgy (*e.g., hardening, tempering, annealing, quenching*) and the mechanical properties of materials (*e.g., tensile strength, hardness, ductility, malleability, elasticity*).

B3. Metrology Skills

By the end of this course, students will:

- B3.1** use the proper measuring tools (*e.g., vernier caliper, inside/outside/depth micrometers, dial indicators, height gauges, gauge blocks, sine bars*) to precisely measure objects to a specified tolerance, and calibrate measuring tools using acceptable national standards;
- B3.2** apply advanced layout processes when preparing a component or project for production;
- B3.3** apply an inspection technique (*e.g., statistical process control*) to evaluate projects and ensure quality.

B4. Selection and Use of Tools and Equipment

By the end of this course, students will:

- B4.1** select and use the correct machine tools, hand tools, and power tools (*e.g., wrenches, electric drills, grinders, rollers, cutters, box and pan brake, shears, engine lathe, milling machine*) when manufacturing a project;
- B4.2** use the appropriate tools and/or equipment for various assembly techniques (*e.g., using fasteners, using adhesives, riveting, soldering, welding, brazing, silver soldering, tinning, using jigs, using fixtures*);
- B4.3** adapt machine tools safely to allow for a variety of uses (*e.g., grinding, drilling, cutting*);
- B4.4** develop and apply a preventive maintenance schedule for the manufacturing shop (*e.g., detailing proper storage techniques, tool and equipment maintenance procedures*).

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of the importance of using sustainable and environmentally friendly manufacturing practices;
- C2.** explain how the manufacturing industry affects society locally, provincially, and/or nationally.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** demonstrate an understanding and application of the three Rs in a manufacturing facility (*e.g., reduction of waste through efficient selection and conversion of materials, reuse of materials when possible, effective collection and recycling of materials and/or fluids*);
- C1.2** demonstrate the use of proper techniques for the disposal of obsolete and/or waste products;
- C1.3** describe the advantages and disadvantages of using various renewable and sustainable energy sources (*e.g., solar, hydrogen fuel cell, wind, geothermal, tidal*) and alternative fuels (*e.g., biodiesel, ethanol*) in manufacturing;
- C1.4** follow environmentally responsible practices during the design and manufacture of a product (*e.g., minimize waste, consider using renewable or recyclable materials, design and manufacture products that last or can be repaired as opposed to throw-away products, use processes that have minimal impact on workers and the local environment*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** describe the effect the manufacturing industry has on the local community and explain the advantages of manufacturing locally;
- C2.2** explain how the globalization of manufacturing industries affects Canadian society locally, provincially, and/or nationally (*e.g., explain the effects of trade agreements, worker health and safety standards or the lack of such standards, environmental standards or the lack of such standards*);
- C2.3** demonstrate an understanding of the role and value of skilled trades in Canadian society (*e.g., in terms of having a skilled and competitive workforce, employment opportunities, services provided*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and compliance with health and safety legislation, standards, and practices, including methods to address deficiencies, as they relate to the manufacturing industry;
- D2.** demonstrate an understanding of the postsecondary pathways leading to careers in manufacturing and the training and certification required for these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** demonstrate an understanding of the specific components of legislation and standards related to workplace safety in the manufacturing industry (e.g., *Occupational Health and Safety Act [OHSA]*, *Workplace Hazardous Materials Information System [WHMIS]*) and the specific responsibilities of the relevant oversight/regulatory organizations (e.g., *Workplace Safety and Insurance Board [WSIB]*, *Industrial Accident Prevention Association [IAPA]*, *Construction Safety Association of Ontario [CSAO]*);
- D1.2** identify and list key ways to promote safe and productive work practices in the manufacturing workplace (e.g., *develop and use a safety checklist for work practices, tools, equipment, and operations; develop and conduct safety audits and inspections of the school manufacturing facility*);
- D1.3** use proper ventilation and/or filtration systems to control air quality (e.g., *to minimize the effects of welding fumes, plastic off-gassing, cutting-fluid misting, and heat treating*);
- D1.4** handle materials safely and appropriately in compliance with the information included in the Material Safety Data Sheets (MSDS) from the Workplace Hazardous Materials Information System (WHMIS);
- D1.5** describe how ergonomics affects productivity, product quality, cost savings/efficiencies, and employee needs and satisfaction, and identify ergonomic considerations related to workshop layout and set-up (e.g., *material handling, ease of movement, lighting, workstation design*);
- D1.6** use and maintain protective clothing and equipment as required to ensure their own and others' safety in the work environment.

D2. Career Opportunities

By the end of this course, students will:

- D2.1** evaluate the career opportunities available through apprenticeships, other training programs, and direct entry into the workforce, and assess their personal aptitude for such opportunities;
- D2.2** research and report on opportunities available locally to obtain certifications and/or training that it would be beneficial to have if they are pursuing a career in manufacturing (e.g., *cardio-pulmonary resuscitation [CPR]*, *Standard First Aid*, *propane safety training*, *equipment lockout training*, *confined space training*, *Canadian Welding Bureau certification*, *fall protection training*);
- D2.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (e.g., *decision making*, *job task planning and organizing*, *problem solving*);
- D2.4** demonstrate an understanding of and apply the work habits that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (e.g., *working safely*, *teamwork*, *initiative*, *working independently*);
- D2.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in manufacturing technology (e.g., *Passport to Safety certificate*, *Ontario Skills Passport Work Plan and Transition Plan*, *photographs of a project*, *technical drawings*, *reports*, *assignments*), and explain why having a current portfolio is important for career development and advancement.